Rohit Ahire

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Professional Summary

Master's student in Data Science & AI at Universität des Saarlandes and FGVT Research Assistant focused on computer vision and AI automation. Experienced with NLP, deep learning, PyTorch, Transformers, FAISS, and FastAPI.

Education

Universität des Saarlandes, MSc in Data Science and Artificial Intelligence *Oct 2024 – Present*
Pune University, India, BE in Information Technology *June 2020 – June 2024*

Experience

Student Research Assistant (HiWi), FGVT, htw saar, Germany *Oct 2026 – Present*

- Developing a camera-based monitoring workflow for FDM 3D printers to detect spaghetti failures, layer shifts, stringing, and print detachment.
- Evaluating defect datasets, annotation strategies, and computer-vision approaches for real-time print-failure detection.
- Configuring Raspberry Pi and camera capture, and designing a browser-accessible monitoring interface; supporting applied AI automation research.

NLP Intern, LinguaTech Solutions *Apr 2024 – May 2025*

- Fine-tuned BERT and RoBERTa models for text classification and sentiment analysis, improving F1-scores by up to 8%.
- Implemented an NER pipeline using spaCy + Hugging Face to extract entities from 50K+ unstructured documents.
- Developed a semantic similarity system using Sentence Transformers, achieving 95%+ top-1 match accuracy on internal validation queries.
- Deployed ML models as FastAPI microservices, achieving under 120 ms inference latency with seamless integration into internal tools.

Machine Learning Intern, Toward's Imagination Pvt. Ltd., India *Jan 2023 – Oct 2023*

- Processed and engineered 3M+ OHLCV time-series records from multiple stock exchanges to create training-ready datasets.
- Trained and benchmarked XGBoost, LSTM, and GRU models, improving short-term trend prediction accuracy by 12% over baseline.
- Engineered 10+ technical indicator features (RSI, MACD, EMA) and evaluated their impact on model performance.
- Designed a modular backtesting framework to assess strategy robustness, reducing manual evaluation time by 40%.

Projects

CompanyDocs RAG Assistant [GitHub]

- Built a local RAG chatbot over 20 GitLab handbook/docs pages; chunked into 206 segments (500 words, 80 overlap) and served via Gradio with citations.
- Implemented semantic retrieval using Sentence-Transformers (all-MiniLM-L6-v2, 384-d) + FAISS (IndexFlatIP) over 206 vectors; returns top-4 evidence with similarity scores.
- Evaluated on 35 questions (25 supported / 10 unsupported): top-4 accuracy 0.56, top-1 0.28, avg retrieval 0.0109s, unsupported rejection 0.80; beat keyword baseline by +20.0 pts (0.36 → 0.56) and cut latency 3.2× (0.0349s → 0.011s).

Lane Detection System [GitHub]

- Built a real-time lane detection benchmarking system using PyTorch on 3,626 TuSimple annotated road images, comparing U-Net, CLRNet-style, LaneATT-style, and SCNN-style segmentation models.
- Implemented an end-to-end pipeline for TuSimple annotation parsing, binary lane-mask generation, model training, evaluation, and visual prediction overlays.
- Achieved best performance with a lightweight U-Net model: 0.389 IoU, 0.559 F1-score, and 91.5 FPS inference, suitable for real-time lane detection.
- Designed and evaluated multiple segmentation architectures using metrics including IoU, F1-score, precision, recall, latency, FPS, parameter count, and model size.
- Developed a local Gradio demo for image upload, model selection, and lane overlay visualization, making the project reproducible and interview-ready.

Heart Disease Risk Prediction Model [GitHub]

- Built an end-to-end clinical risk prediction pipeline using stratified CV, balanced preprocessing, and hyperparameter-tuned XGBoost.
- Achieved 0.90 ROC-AUC and improved recall using threshold calibration.
- Applied SHAP for patient-level and global interpretability, identifying key factors such as thal, oldpeak, and ca.

Technical Skills

Programming: Python, PyTorch, TensorFlow, Keras, Scikit-learn
NLP: Transformers, spaCy, Sentence Transformers, BERT, RoBERTa, LLaMA
Computer Vision: CNNs, U-Net, SCNN, LaneATT, CLRNet
Tools: FastAPI, Git, Jupyter, VS Code, Cursor AI, Codex